

07 The Case Against Generation

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10 MIN READ · INTERACTIVE

When the future is genuinely open, the best possible pixel prediction is a picture of something that cannot happen. The case for predicting descriptions instead, and the bill that arrives with it.

Point a camera at a road and ask what the next second looks like.

A leaf on a tree at the edge of the frame will move. Which way it moves is unknowable, and it is also the least interesting fact available. Nothing you might decide to do depends on it.

But a model asked to predict the picture has to predict that leaf, because the leaf is made of pixels and pixels are what it is being marked on. Multiply that by every leaf, every ripple, every flicker of light on wet tarmac. Most of the capacity in the system has now gone to things that are hard to predict and do not matter.

That is the smaller of the two problems.

The average of two futures is neither of them

Here is the larger one.

A car is coming to a junction. It will turn left or it will turn right. You genuinely do not know which, and no amount of looking harder at the current frame will tell you.

Ask a model to predict the next picture and score it by how close the pixels are, and there is a mathematically best answer. It is not either turn. It is the average of them.

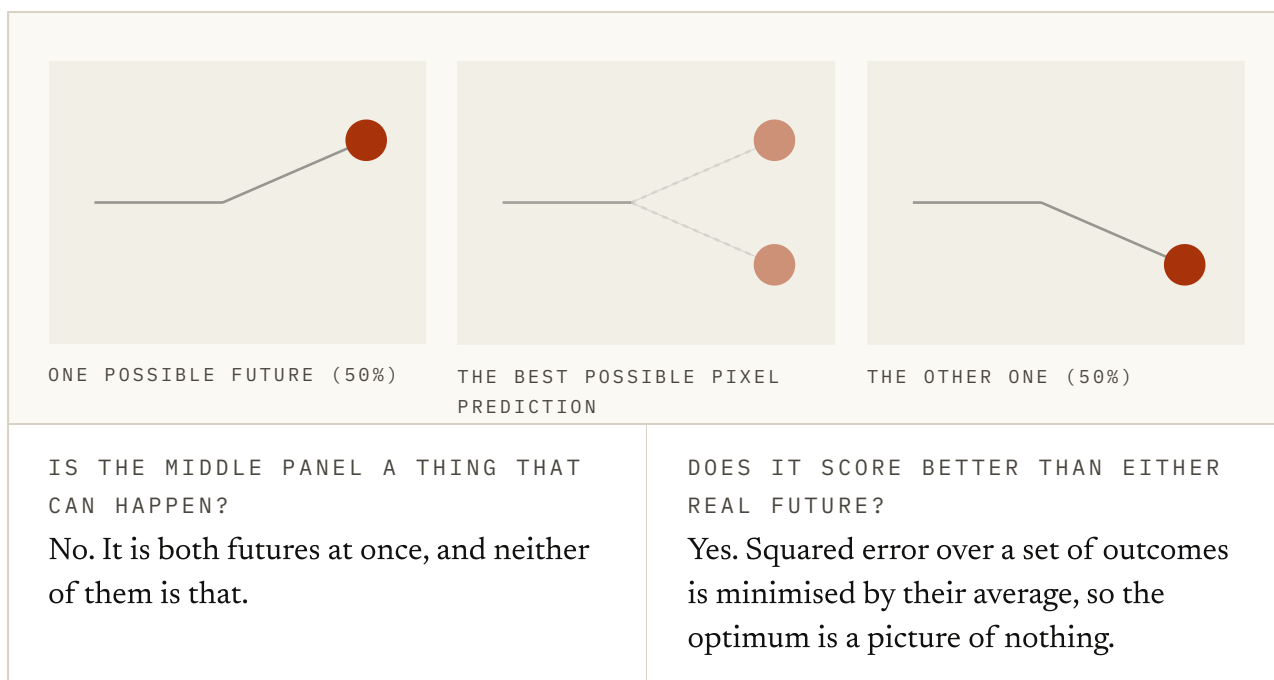


FIG. 7.1 The middle panel is not an impression of blurriness. It is the two futures drawn at their probabilities, which is their average, which is exactly what squared error is minimised by. Slide the chance towards either end and the optimum resolves; leave it near the middle and the best available prediction is a picture of something that cannot happen.

Read that middle panel again. It scores better than either of the real futures. A model that confidently predicted the left turn would be punished half the time; the ghost is punished a little all the time, and a little all the time wins.

So the training signal is not asking for a plausible future. It is asking for the mean of the futures, and when the future is genuinely open, the mean is a smear that never occurs.

This is why generated video gets vague exactly when something interesting is about to happen. Uncertainty is where the smear comes from, and uncertainty is concentrated at the moments that matter.

Predict the description instead

The proposal is short. Stop predicting the picture.

Encode what you can see into a compact description (an *embedding*: a short list of numbers standing in for something bigger, arranged so that similar things land near each other). Encode the part you are predicting into one too. Then train the model to predict the second description from the first, and never decode anything back into pixels at all.

This buys two things at once.

The leaf problem goes away, because the description was never obliged to record which way the leaf went. If a detail does not survive the encoding, nothing downstream is graded on it.

The averaging problem gets smaller, because you are no longer averaging pictures. The description of a car turning left and the description of a car turning right can share almost everything. What is being predicted may be a single point that both of them are close to, so the prediction has somewhere sensible to land.

This is the JEPA family, and the name says the architecture: joint embedding, because both sides get embedded, and predictive, because the training signal is one embedding predicting another.

The bill for that

Now the part that is usually skipped, and it is not a small part.

Predicting pixels has one enormous virtue. The target is fixed. The next frame is the next frame, and the model cannot make it easier.

Predict an embedding and that stops being true, because the thing you are predicting is produced by a network that is also being trained. There is now a way to get a perfect score that involves learning nothing at all: make every description identical. The prediction is then always exactly right, the loss goes to zero, and the representation says nothing about anything.

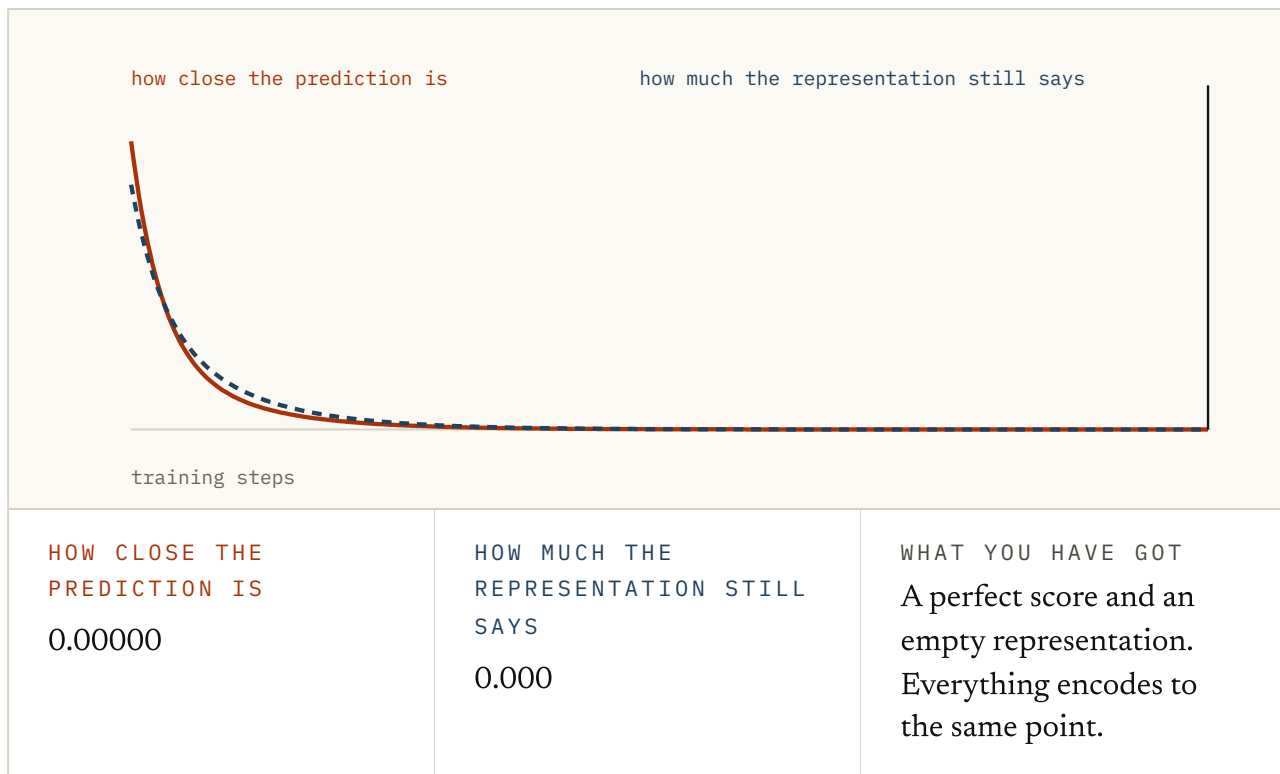


FIG. 7.2 A two-by-two encoder trained on four pairs by gradient descent. Nothing here is illustrative: the numbers are whatever the arithmetic produces. Without the safeguard the loss reaches zero and the representation reaches zero with it. Turn the safeguard on and the loss never gets close to zero, which is the point. The useful run is the one that looks worse.

Everything difficult about these methods lives in preventing that. Keep a second copy of the encoder that updates slowly (usually called a *target encoder*: it trails the one being trained, so the thing being predicted cannot sprint away from the thing predicting it), so the target cannot run away. Push different inputs apart explicitly. Penalise descriptions whose components all say the same thing. There are several families of answer and none of them is elegant.

Notice what has happened to the loss. It is no longer a measurement you can read. A pixel loss of zero means the model predicted the frame. An embedding loss of zero might mean it understands everything, or nothing at all, and the number will not tell you which. The score stopped being trustworthy at the same moment it stopped being about pixels.

What this does not settle

It would be tidy to end with generation being a mistake. It is not.

Systems that predict pixels are the ones producing worlds you can walk around in, and they work. Whatever is being wasted on leaves, they are delivering something that no amount of embedding prediction delivers, which is an actual picture. If you want to see the future rather than reason about it, somebody has to draw it.

So the honest statement of the argument is narrower than the slogan. Predicting appearances spends most of its capacity on things no decision depends on. And when the future is uncertain, it is trained towards an image of something that will not happen. If what you want is a compact state to act on, that is a bad way to get one. If what you want is a picture, it is the only way anyone has.

The two goals were never the same goal. Most of the confusion in this corner of the field is people comparing systems that were built for different ones.

Try it

1 A car will turn left or right with equal probability. What does a pixel loss consider the best prediction?

- a. The left turn
- b. The right turn
- c. The average of the two, which is a picture of neither
- d. Whichever was more common in training

ANSWER c. The average of the two, which is a picture of neither. Squared error over a set of outcomes is minimised by their mean. Committing to one turn is punished half the time; the smear is punished a little all the time, and that wins.

2 Why does generated video tend to get vague exactly when something interesting is about to happen?

- a. The models run out of capacity
- b. Interesting moments are where the future is uncertain, and uncertainty is what produces the smear
- c. Frame rates drop
- d. The encoder is too small

ANSWER b. Interesting moments are where the future is uncertain, and uncertainty is what produces the smear. The vagueness is not a rendering problem. It is the training objective asking for the mean of the possible futures, and that mean is least like any one of them where the outcome is most open.

3 A leaf at the edge of the frame moves unpredictably. Why does a pixel-predicting model spend capacity on it?

- a. Leaves are important for scene understanding
- b. The leaf is made of pixels, and pixels are what the model is marked on
- c. It cannot tell leaves from cars
- d. The encoder forces it to

ANSWER b. The leaf is made of pixels, and pixels are what the model is marked on. Nothing tells the objective which pixels matter. Hard-to-predict and irrelevant is the worst combination, and it is most of the frame.

4 What does predicting an embedding rather than a picture do to the leaf problem?

- a. It makes the leaf easier to predict
- b. If a detail does not survive the encoding, nothing downstream is graded on it
- c. It moves the problem to the decoder
- d. Nothing, the leaf is still in the input

ANSWER b. If a detail does not survive the encoding, nothing downstream is graded on it. The description was never obliged to record which way the leaf went, so the model is not punished for failing to say.

5 Predicting pixels has one large virtue that predicting embeddings gives up. What is it?

- a. It is faster to compute
- b. The target is fixed: the model cannot make the next frame easier
- c. It needs less data
- d. It generalises better

ANSWER b. The target is fixed: the model cannot make the next frame easier. The moment the target is produced by a network that is also being trained, the target can move, and there is a way to make it move somewhere very convenient.

6 What is collapse?

- a. The loss diverging to infinity
- b. Every input encoding to the same description, so the prediction is always right and the representation says nothing
- c. The model forgetting earlier training
- d. Gradients vanishing in deep layers

ANSWER b. Every input encoding to the same description, so the prediction is always right and the representation says nothing. It is a perfect score obtained by learning nothing, and it is the cheapest solution available unless something is specifically stopping it.

7 In Figure 7.2, the run with the safeguard has a worse loss. What does that tell you?

- a. The safeguard is badly tuned
- b. An embedding loss is no longer a number you can read off as quality
- c. The model needs more training
- d. The safeguard should be removed once training stabilises

ANSWER b. An embedding loss is no longer a number you can read off as quality. A pixel loss of zero means the frame was predicted. An embedding loss of zero might mean everything or nothing, and the number cannot tell you which.

8 Does this argument show that generating pixels is a mistake?

- a. Yes, it is strictly worse
- b. No. It is a bad way to get a compact state to act on, and the only way anyone has to get an actual picture
- c. Yes, except for very short videos
- d. No, because collapse makes embeddings unusable

ANSWER b. No. It is a bad way to get a compact state to act on, and the only way anyone has to get an actual picture. The two goals were never the same goal, and most of the confusion here is people comparing systems built for different ones.

FIG. 7.3 Eight questions on the argument and its bill. The last two are the ones that stop this reading as a knock-down case.

Which leaves the systems that went the other way, and went hard.

Thanks for reading. Chapter 8 is about what happens when you scale pixel prediction until the result is something you can steer with a keyboard. Also about what "it learned physics" does and does not mean when someone says it about a video model.

Sources

A Path Towards Autonomous Machine Intelligence LECUN, 2022 ↗

The position paper the whole argument comes from, including why predicting appearances is the wrong job for a system meant to act.

<https://openreview.net/forum?id=BZ5a1r-kVsF>

I-JEPA ASSRAN ET AL., 2023 ↗

Hide part of an image and predict the embedding of the missing piece rather than redrawing it. The clean statement of the method.

<https://arxiv.org/abs/2301.08243>

V-JEPA 2 META AI, 2025 ↗

The video version, pre-trained without actions and then post-trained for control, which is where the argument meets a robot.

<https://ai.meta.com/blog/v-jepa-2-world-model-benchmarks/>

Bootstrap Your Own Latent (BYOL) GRILL ET AL., 2020 ↗

The slowly-updating target copy, and the result that made people believe you could avoid collapse without pushing things apart.

<https://arxiv.org/abs/2006.07733>

VICReg BARDES, PONCE & LECUN, 2021 ↗

The explicit approach: penalise a representation whose components have collapsed or duplicated each other. Figure 7.2's safeguard, done properly.

<https://arxiv.org/abs/2105.04906>

A Cookbook of Self-Supervised Learning BALESTRIERO ET AL., 2023 ↗

The honest survey, including how much of this field is machinery for stopping the trivial solution from winning.

<https://arxiv.org/abs/2304.12210>

Masked Autoencoders Are Scalable Vision Learners HE ET AL., 2021 ↗

The counter-example worth holding on to: reconstruct the pixels of the masked part, and it works very well anyway.

<https://arxiv.org/abs/2111.06377>

Learning and Leveraging World Models in Visual Representation Learning

GARRIDO ET AL., 2024 ↗

What the embedding-prediction objective turns out to have learned, tested rather than asserted.

<https://arxiv.org/abs/2404.08471>

FIG. 7.4 Ordered for someone building on this rather than for historical completeness.